

Pramo Jewels

Database Design Document

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Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define the relational database architecture, entities, relationships and standards
Audience	Backend Developers, DBAs, Architects, QA
Version	1.0

Core Entities

Table	Purpose
users	Customer/admin accounts
roles	RBAC roles
addresses	Shipping & billing
categories	Product categories
products	Jewellery catalogue
product_images	Gallery images
inventory	Stock management
wishlists	Saved items
carts	Shopping carts
cart_items	Cart products
orders	Customer orders
order_items	Purchased products
payments	Payment records
shipments	Courier tracking
reviews	Ratings & reviews
coupons	Discount campaigns
notifications	System notifications
audit_logs	Security & admin logs

ERD Overview

Primary relationships include: Users→Orders (1:N), Orders→Order Items (1:N), Products→Order Items (1:N), Categories→Products (1:N), Users→Addresses (1:N), Products→Reviews (1:N), Users→Reviews (1:N), Orders→Payments (1:1), Orders→Shipments (1:1).

Schema Standards

All tables use bigint/UUID identifiers, created_at, updated_at and soft deletes where appropriate. Foreign keys enforce referential integrity.

Indexing Strategy

Create indexes for email, SKU, category_id, order_number, payment_reference, shipment_tracking_id and frequently filtered fields.

Constraints

Use unique constraints for email, SKU and coupon codes. Apply check constraints for positive prices and quantities where supported.

Normalization

Maintain at least Third Normal Form (3NF). Store derived values only when required for reporting or performance.

Security

Encrypt sensitive fields, hash passwords, never store raw payment credentials, and audit privileged operations.

Performance

Use pagination, eager loading, selective indexing and Redis caching for read-heavy queries.

Migration Standards

Implement Laravel migrations, foreign key constraints, seeders and factories. All schema changes must be version controlled.

Future Expansion

Reserve extensibility for loyalty points, AI recommendations, multilingual content, gift cards and mobile applications.